

Tide retrospective: sampler laws for a Gaussian target the ULA closed form, the minibatch Lyapunov entries, the finite-time identity and the Gaussian LLC

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Abstract

Let P be a real symmetric positive definite precision matrix with eigenvalues p_i and let $h > 0$ satisfy $h p_i < 2$ for every i . The Euler-discretised Langevin chain $w \leftarrow (I - hP)w + \sqrt{2h}\xi$ has a unique stationary covariance, namely $\Sigma_{\text{ULA}} = (P - \frac{h}{2}P^2)^{-1}$; it is positive definite, commutes with P , and has eigenvalues $1/(p_i(1 - hp_i/2))$. If the noise covariance is instead $2hI + h^2t^2C$ for an arbitrary matrix C , the unique stationary covariance has entries $(2h\delta_{ij} + h^2t^2\tilde{C}_{ij})/(h(p_i + p_j) - h^2p_ip_j)$ in the eigenbasis of P , whose diagonal is the ULA variance times $1 + ht^2\tilde{C}_{ii}/2$. The iterates of the covariance recursion from any start satisfy $\Sigma_k - \Sigma_\infty = A^k(\Sigma_0 - \Sigma_\infty)(A^\top)^k$, which gives $\Sigma_\infty(I - A^{2k})$ from a zero start and $\sigma^2(1 - \rho^{2k})$ in one dimension. On the Gibbs side, in the seabed's own language, the expectation of $K = \frac{1}{2}w^\top Hw$ under the weight $\exp(-tK - \gamma|w|^2/2)$ is $\frac{1}{2} \text{tr}(H(tH + \gamma I)^{-1})$, and $t\langle K \rangle = d/2$ when $\gamma = 0$. These are the closed forms that the note *Sanity on Sampling* checks stochastic gradient Langevin dynamics against. Three new modules, 715 lines, no **sorry**, **axiom** or **native_decide**; one deliberation round in which GPT-6 Astra strengthened two of the three candidates; a single session.

1 Setting

The note *Sanity on Sampling* (paper **sanity** in the SRI registry, Overleaf clone `learning-theory/papers/sanity`, code in `timaeus-research/sanity-sampling`) compares two ways of computing posterior covariances and the local learning coefficient for nondegenerate potentials: the inverse Hessian, through the Laplace formulas of the Susceptibility Primer, and sampling with SGLD. Its experiments E1 to E8 rest on a small set of closed forms for the Gaussian linear recursion: the ULA law (its equation `eq:ula`), the Lyapunov law with minibatch noise (`eq:mblaw`), the AR(1) variance of a finite chain, and the Gaussian identities for the localised LLC (`eq:llc`). All of them were derived by hand in the note and checked numerically; none had been formalised.

The seabed is the `laplace` repository at commit `b7e20dd`. It carries the Gibbs objects (`partitionFunction`, `gibbsExpectation`, `gibbsCov`), the multivariate Gaussian weight `gaussianWeight`, the quadratic form `quadForm`, and the Gaussian integration by parts theorem `gaussian_second_moment_eq_inverse_entry`, which says that the second moment of $e^{-\frac{1}{2}u^\top Pu}$ is $Z(P^{-1})_{ij}$. It has nothing on discrete-time samplers, Markov chains or Lyapunov equations, so this tide opens a new arc, `Laplace/Sampler/`.

The deliberation is recorded in `tide-log/`: the log entry carries the same date stamp as this retrospective, and the consult is `gpt_sampler_laws_v1.md`. It started from three candidates: the Lyapunov laws as linear algebra, the Gaussian invariance of the ULA step as a probabilistic statement, and the Gaussian LLC identity in the seabed's language. GPT-6 Astra confirmed all three, deferred the probabilistic one as measure-theoretic plumbing, and strengthened the other two: the minibatch formula needs no commutation between C and P , and the finite-time identity holds for any starting covariance without any commutation. It also fixed the route:

GPT-6 Astra: Route: spectral, not Neumann. Diagonal Lyapunov solver $X_{ij} = N_{ij}/(1 - a_i a_j)$ for $|a_i| < 1$, transported through an orthogonal diagonalisation; existence and uniqueness become scalar algebra. State the stability hypothesis eigenvalue-wise: the sup norm on $\iota \rightarrow \mathbb{R}$ does not give $\|I - hP\|_2 < 1$.

Both parties voted for the six-item cluster after one round.

2 The thing formalised

Write $A = I - hP$ for the ULA step, $\mathcal{T}_N(X) = AXA^\top + N$ for the covariance step, and U for an orthogonal matrix with $P = U \text{diag}(p) U^\top$.

- (L) For $a \in \mathbb{R}^d$ with $|a_i| < 1$ and any N , the equation $X = \text{diag}(a) X \text{diag}(a) + N$ has the unique solution $X_{ij} = N_{ij}/(1 - a_i a_j)$. For $A = U \text{diag}(a) U^\top$ with U orthogonal, $X = AXA^\top + N$ has the unique solution $U [(U^\top N U)_{ij}/(1 - a_i a_j)] U^\top$.
- (U) For P symmetric positive definite and $0 < h$ with $hp_i < 2$: $\Sigma_{\text{ULA}} = (P - \frac{h}{2}P^2)^{-1}$ is the unique solution of $\Sigma = A\Sigma A + 2hI$, it is positive definite, and $(U^\top \Sigma_{\text{ULA}} U)_{ij} = \delta_{ij}/(p_i(1 - hp_i/2))$.
- (M) With $N = 2hI + h^2 t^2 C$ and $\tilde{C} = U^\top C U$, the unique solution has $(U^\top \Sigma U)_{ij} = (2h\delta_{ij} + h^2 t^2 \tilde{C}_{ij})/(h(p_i + p_j) - h^2 p_i p_j)$, and on the diagonal $(1 + ht^2 \tilde{C}_{ii}/2)/(p_i(1 - hp_i/2))$.
- (F) If S is a fixed point of $\mathcal{T}_N$ then $\mathcal{T}_N^k(X_0) - S = A^k(X_0 - S)(A^\top)^k$; for $X_0 = 0$, A symmetric and $AS = SA$ this is $S(I - A^{2k})$; in one dimension the variance after k steps is $\frac{v}{1-\rho^2}(1 - \rho^{2k})$.
- (G) For $P = tH + \gamma I$ with the seabed's integrability package,

$$\frac{1}{Z(P)} \int \frac{1}{2} \langle u, H u \rangle e^{-\frac{1}{2} \langle u, P u \rangle} du = \frac{1}{2} \sum_{ij} H_{ij} (P^{-1})_{ij},$$

and for symmetric H and $\gamma = 0$ the right side is $d/(2t)$.

The Lean signatures of (U) and (M), transliterated (U is `orthoOf hP`, the real orthogonal matrix packaged from Mathlib's `eigenvectorUnitary`):

```
theorem ula_fixed_iff {P : Matrix i i R} (hP : P.IsHermitian) (h : R)
  (hh : 0 < h)
```

```

(hev : forall i, 0 < hP.eigenvalues i and h * hP.eigenvalues i < 2) (X) :
covStep (ulaStep P h) (ulaNoise h) X = X <->
  X = lyapunovVia (orthoOf hP) (fun i => 1 - h * hP.eigenvalues i)
    (ulaNoise h)

theorem ulaCov_fixed (P : Matrix i i R) (h : R) (hP : P^T = P)
(hdet : IsUnit (P - (h / 2) * (P * P)).det) :
covStep (ulaStep P h) (ulaNoise h) (ulaCov P h) = ulaCov P h

theorem ulaCov_posDef {P} (hP : P.PosDef) (h : R) (hh : 0 < h)
(hev : forall i, h * hP.1.eigenvalues i < 2) : (ulaCov P h).PosDef

theorem minibatchCov_conj_apply {P} (hP : P.IsHermitian) (h t : R) (C) (i j) :
((orthoOf hP)^T * lyapunovVia (orthoOf hP)
  (fun i => 1 - h * hP.eigenvalues i) (minibatchNoise h t C)
  * orthoOf hP) i j =
((2 * h) * (if i = j then 1 else 0)
  + h ^ 2 * t ^ 2 * ((orthoOf hP)^T * C * orthoOf hP) i j) /
(h * (hP.eigenvalues i + hP.eigenvalues j)
  - h ^ 2 * hP.eigenvalues i * hP.eigenvalues j)

```

The other declarations, by module:

- Laplace/Sampler/Lyapunov.lean: (L) as diagLyapunov_fixed, diagLyapunov_unique, covStep_diagonal_fixed_iff, lyapunovVia_fixed_iff, lyapunovVia_conj_apply; the real spectral packaging orthoOf, spectral_real, symmLyapunov_fixed_iff; (F) as covStep_iterate_sub_fixed, covStep_iterate_zero, covStep_iterate_zero_of_comm, ar1_var_iterate.
- Laplace/Sampler/ULA.lean: (U) as ulaCov_commute, ulaCov_fixed, isUnit_det_ulaDenom, ulaCov_eq_lyapunov, ulaCov_conj_apply, ulaCov_posDef; (M) as minibatchCov_conj_apply, minibatchCov_conj_diag, minibatch_fixed_iff.
- Laplace/Multi/GaussianLLC.lean: (G) as gaussian_quadForm_integral, localised_gaussian_K_expectation, gaussian_K_expectation_eq_half_dim, with the pairing hessInvPairing $H \text{ Pinv} = \sum_{ij} H_{ij}(P^{-1})_{ij}$.

In the note, (U) is the ULA law, the black curve of experiment E1 and the correction applied throughout; (M) is the Lyapunov law of E8, whose diagonal is the “ $1 + \text{lr } t^2 c_i / 2$ ” inflation of the per-direction variance; (F) is the finite-chain correction of E2 and E4; (G) is the Hessian value of the localised LLC that E3 and E6 compare against. The stability hypothesis $hp_i < 2$ is the note’s $\text{lr } p_{\max} < 2$. The hypothesis package of (G) is the seabed’s: integrability of the weight and its second moments, and the Fubini hypothesis of the integration by parts theorem, all stated abstractly rather than derived from positive definiteness of P .

3 Proof strategy

Diagonal solver and transport. For $D = \text{diag}(a)$ the covariance step acts entrywise, $(DXD + N)_{ij} = a_i a_j X_{ij} + N_{ij}$, so a fixed point is a scalar equation per entry with the unique solution $N_{ij} / (1 - a_i a_j)$ because $1 - a_i a_j > 0$ when $|a_i|, |a_j| < 1$. Conjugation by an

orthogonal U intertwines the steps, $U^\top \mathcal{T}_{A,N}(X)U = \mathcal{T}_{D,U^\top NU}(U^\top XU)$ (`covStep_conj`), and is a bijection on matrices, so the fixed points of the general step are the transports of the diagonal ones. This is `lyapunovVia_fixed_iff`. Mathlib’s spectral theorem for Hermitian matrices produces a unitary over $\mathbb{R}$; `orthoOf` coerces it to a real matrix and `spectral_real` restates $A = U \text{diag}(\lambda) U^\top$ using `conjTranspose_eq_transpose_of_trivial` to turn the star into a transpose.

ULA closed form. The step satisfies $I - A^2 = 2hP(I - \frac{h}{2}P) = 2h(P - \frac{h}{2}P^2)$, a two-line module identity (`one_sub_ulaStep_sq`). Since Σ_{ULA} is the inverse of a polynomial in P it commutes with P (`ulaCov_commute`, from `mul_nonsing_inv` and `nonsing_inv_mul`), hence with A , so $\Sigma - A\Sigma A = (I - A^2)\Sigma = 2h(P - \frac{h}{2}P^2)\Sigma = 2hI$. This is `ulaCov_fixed`; it needs only $P^\top = P$ and the invertibility of $P - \frac{h}{2}P^2$. Under the stability hypothesis the determinant is $\prod_i p_i \prod_i (1 - hp_i/2) \neq 0$ (`isUnit_det_ulaDenom`, through `det_eq_prod_eigenvalues` and `ulaStep_eq_conj`), so uniqueness follows from `lyapunovVia_fixed_iff`, and the eigenbasis entries are $2h/(1 - (1 - hp_i)^2) = 1/(p_i(1 - hp_i/2))$. Positivity is `PosDef.diagonal` for the diagonal solution and `PosDef.conjTranspose_mul_mul_same` for its transport.

Minibatch entries. With $N = 2hI + h^2 t^2 C$ the transported formula gives $(2h\delta_{ij} + h^2 t^2 \tilde{C}_{ij})/(1 - (1 - hp_i)(1 - hp_j))$, and the denominator is $h(p_i + p_j) - h^2 p_i p_j$ by `ring`. No commutation is used anywhere; the diagonal specialisation is a `div_eq_div_iff` plus `ring`.

Finite time. $\mathcal{T}_N(Y) - S = A(Y - S)A^\top$ for a fixed point S , so induction with `Function.iterate_succ_apply'` gives $A^k(X_0 - S)(A^\top)^k$. With $X_0 = 0$, $A = A^\top$ and $AS = SA$ the right side is $-SA^{2k}$, whence $S(I - A^{2k})$; the scalar `AR(1)` case is the same induction on $\mathbb{R}$.

Gaussian LLC. Writing $u = \sum_j u_j e_j$ (the seabed’s `eq_sum_stdBasis`) gives $\langle u, Hu \rangle = \sum_{ij} H_{ij} u_i u_j$; the integral of each term against the Gaussian weight is $Z(P^{-1})_{ij}$ by `gaussian_second_moment_eq_inverse_entry`, and the finite sums pass through the integral by `integral_finsetSum`. Dividing by Z gives $\frac{1}{2} \sum_{ij} H_{ij}(P^{-1})_{ij}$. For $P = tH$ and $P^{-1} = \frac{1}{t}H^{-1}$ the pairing collapses: $\sum_{ij} H_{ij}(H^{-1})_{ij} = \sum_{ij} H_{ji}(H^{-1})_{ij} = \text{tr}(HH^{-1}) = d$ using the symmetry of H , so $t\langle K \rangle = d/2$.

4 Roadblocks and resolutions

None of the mathematics resisted; the friction was all Lean idiom, and most of it in the spectral packaging.

- Mathlib’s `Matrix.IsHermitian.spectral_theorem` now reads

$$A = \text{Unitary.conjStarAlgAut } \mathbb{R} _ U \left(\text{diag}(\text{RCLike.ofReal} \circ \lambda) \right).$$

Unfolding needs `Unitary.conjStarAlgAut_apply` (the `Matrix.`-prefixed name no longer exists), and the `RCLike.ofReal` over $\mathbb{R}$ is closed by `congr 1` alone; adding a `funext` or `simp` after it errors with “no goals”. The same “no goals” appeared twice more when `congr 1` finished a goal that I had followed with `ring` bullets.

- The diagonal solver’s uniqueness reduced to $x = n/(1 - a^2)$ from $x = a^2 x + n$; `field_simp` followed by `linear_combination` left a residue, and `rw [eq_div_iff hpos.ne']`

followed by `linear_combination -h` closed it.

- In the commuting finite-time identity, rewriting $S A^{2k}$ into $S A^k A^k$ failed on associativity until the chain was ordered `mul_sub`, `mul_one`, `hc`, `mul_assoc`, `← pow_add`, `two_mul`.
- `simp` only `[quadForm, clm_apply_eq_sum, Finset.mul_sum]` loops: the expansion of $(Hu)_i$ produces terms $(He_j)_i$ that match the same lemma again. The fix is a targeted `rw` inside `Finset.sum_congr`.
- Scalar cancellation orientation: the goal was $(\frac{1}{t} \cdot t) \bullet (H \circ H^{-1}) = \text{id}$, so `one_div_mul_cancel`, not `mul_one_div_cancel`.
- After `split_ifs` in the diagonal positivity computation the $i = j$ branch still mentioned λ_j ; `subst then rw [mul_one]` closed it.
- Deprecations since the seabed's pin: `integral_finset_sum` is now `integral_finsetSum`, `integrable_finset_sum` is `integrable_finsetSum`, and `ContinuousLinearMap.smul_apply` resolves under the open namespace as `smul_apply`.

The one architectural point worth recording: the deliberation proposed stating stability as $\|A\|_2 < 1$ and noted that Mathlib's operator norm on $\iota \rightarrow \mathbb{R}$ is the sup norm, which does not give it. Stating the hypothesis eigenvalue-wise ($|1 - hp_i| < 1$, equivalently $0 < hp_i < 2$) avoided `EuclideanSpace` entirely, and every theorem is stated on `Matrix  $\iota \iota \mathbb{R}$` with `Fintype` and `DecidableEq` instances on ι .

5 What was Mathlib, what was new

Mathlib, all lookup: `Matrix.IsHermitian.spectral_theorem`, `eigenvectorUnitary`, `eigenvalues`, `Unitary.conjStarAlgAut_apply`, `Matrix.UnitaryGroup.star_mul_self`, `mem_unitaryGroup_iff`, `conjTranspose_eq_transpose_of_trivial`, `IsHermitian.det_eq_prod_eigenvalues`, `PosDef.eigenvalues_pos`, `PosDef.diagonal`, `PosDef.conjTranspose_mul_mul_same`, `det_mul`, `det_diagonal`, `det_transpose`, `isUnit_iff_isUnit_det`, `mul_nonsing_inv`, `nonsing_inv_mul`, the `diagonal_mul/mul_diagonal` family, `Function.iterate_succ_apply'`, `Commute.pow_left`, `integral_finsetSum`, `integral_const_mul`, and the `module` and `linear_combination` tactics.

Seabed: `gaussianWeight`, `gaussianZ`, `quadForm`, `FubiniIBPHypothesis`, `eq_sum_stdBasis`, `gaussian_second_moment_eq_inverse.entry`.

New: the covariance step and the two Lyapunov solvers with their fixed-point characterisations, the real-orthogonal packaging of the spectral theorem, the ULA closed form with its commutation, determinant and positivity lemmas, the minibatch entry formulas, the finite-time identities, and the quadratic-form bridge `hessInvPairing` with its two LLC corollaries. About 715 lines, 51 declarations.

6 Lessons

- For CLAUDE.md: the real spectral theorem recipe (`orthoOf`, `spectral_real`) and the `congr 1` closure of `RCLike.ofReal` $\circ \lambda$ are worth a section; they will recur in any matrix-valued tide. Landed in this commit.
- For the lean-formalisation skill: for linear-algebraic targets, a skeleton of typechecked `sorry` statements written in one pass and then filled module by module worked with no rework; the only statement change during filling was dropping an unused positivity hypothesis from the determinant lemma.
- For the tide skill: the direction was a six-item programme. Taking the sampler-side cluster and one Gibbs-side bridge, and writing the remaining items into the log as deferred, kept the tide to one coherent excursion. The numerical check was free: the note's experiments E1 and E8 had already tested every closed form at machine precision on the exact-gradient cells.
- For the paper: the Lean statements make explicit what the note states loosely, that the minibatch law needs no commutation of C_g with P and that its diagonal inflation factor is exact for any C_g in the eigenbasis of the precision.

7 Follow-ups

- *Gaussian invariance.* $N(0, \Sigma)$ pushed through one ULA step is $N(0, A\Sigma A^\top + 2hI)$, via `isGaussian_map` and `isGaussian_conv` on `EuclideanSpace  $\mathbb{R} \iota$` ; then $N(0, \Sigma_{\text{ULA}})$ is invariant.
- *Expected sample variance.* Temporal cross-covariances $\text{Cov}(x_{r+l}, x_r) = A^l \Sigma_r$ and the note's finite-chain formula for the variance of n consecutive draws (`ar1_expected_sample_variance` in the code).
- *Full state-dependent law.* $\Sigma = A\Sigma A + 2hI + h^2 t^2 (C + c \sum_i (H_i - H) \Sigma (H_i - H))$, needing a contraction hypothesis on the whole operator, not just on A .
- *Monotonicity.* $N_1 \preceq N_2$ implies $\Sigma(N_1) \preceq \Sigma(N_2)$, hence $\Sigma_{\text{mb}} \succeq \Sigma_{\text{ULA}}$; cheap from the entry formula.
- *Operator-norm wrapper.* $\|I - hP\|_2 < 1$ on `EuclideanSpace` equivalent to the eigenvalue-wise hypothesis.
- *Discharging the Gibbs hypotheses.* Derive the integrability and Fubini package of (G) from P positive definite, so that `localised.gaussian_K.expectation` takes `hP : P.PosDef` alone.
- *One-loop coefficient and Rosenbrock.* The t^{-2} covariance correction of experiment E7 and the $d = 2$ Rosenbrock closed forms of E5, both from the direction, both untouched.
- *Paper tags.* Proposed `\leanref` markers for `eq:ula`, `eq:mblaw` and `eq:llc` of the note are staged in `learning-theory/projects/sanity/staging/`; landing them in the Overleaf clone is a separate step.

Acknowledgements

GPT-6 Astra fixed the spectral route, removed the commutation hypothesis from the mini-batch formula, supplied the finite-time identity in its general form, and flagged the sup-norm trap. The seabed's Gaussian integration by parts is the multivariate laplace work of May 2026.